Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A ladder makes an angle of 60° with the ground when placed against a wall. If the foot of the ladder is 2 m away from the wall, then the length of the ladder (in metres) is: [1]

(A) $\frac{4}{\sqrt{3}}$ (B) $4\sqrt{3}$ (C) $2\sqrt{2}$ (D) 4

Q2. If the angle of elevation of a tower from a distance of 100 m from its foot is 60°, then the height of the tower is: [1]

(A) $100\sqrt{3}$ m (B) $\frac{100}{\sqrt{3}}$ m (C) $50\sqrt{3}$ m (D) $\frac{200}{\sqrt{3}}$ m

Q3. The tops of two poles of heights 16 m and 10 m are connected by a wire of length l metres. If the wire makes an angle of 30° with the horizontal, then l is: [1]

(A) 26 m (B) 16 m (C) 12 m (D) 10 m

Q4. If a 1.5 m tall girl stands at a distance of 3 m from a lamp-post and casts a shadow of length 4.5 m on the ground, then the height of the lamp-post is: [1]

(A) 1.5 m (B) 2 m (C) 2.5 m (D) 2.8 m

Q5. **Assertion (A):** If the angles of elevation of the top of a tower from two points at distances of 9 m and 16 m from the base in the same straight line are 30° and 60°, then the height of the tower is 12 m.
Reason (R): When the angles of elevation of the top of a tower from two points on the same side at distances a and b are complementary, the height of the tower is given by $h = \sqrt{ab}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

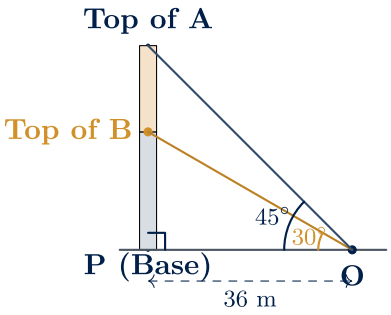
Q6. A kite is flying at a height of 75 m above the ground, attached to a string inclined at 60° to the horizontal. Find the length of the string, assuming there is no slack. [2]

Q7. The angle of elevation of an aeroplane from a point on the ground is 60°. After a flight of 30 seconds, the angle of elevation becomes 30°. If the aeroplane is flying at a constant height of $3000\sqrt{3}$ m, find the speed of the aeroplane in km/h. [3]

Q8. A man on a cliff observes a boat at an angle of depression of 30° approaching the shore immediately beneath the observer with uniform speed. Six minutes later, the angle of depression of the boat is found to be 60°. Find the total time taken by the boat to reach the shore. [3]

Q9. The angles of depression of the top and bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45°, respectively. Find the height of the multi-storeyed building and the distance between the two buildings. [3]

Q10. CASE STUDY: Telecommunication Transmission Tower Engineering



Radio and communication towers are engineered to broadcast high-frequency signals across urban regions. A radio station transmission tower is built in two distinct functional vertical sections on level ground: Section B (lower transmission mast) and Section A (upper microwave relay dish) mounted directly on top of Section B.
The tower is anchored and monitored from an engineering observation station O located on the ground at a horizontal distance of 36 m from the base of the tower (P). From point O, the angle of elevation of the top of Section B is 30° and the angle of elevation of the top of Section A is 45°.

Based on the above mathematical scenario, answer the following questions:

(i) Find the height of the lower transmission mast Section B (i.e., height PB). [1]

(ii) Find the total height of the combined tower from ground to the top of Section A (i.e., height PA). [1]

(iii) Calculate the height of Section A alone (i.e., vertical distance AB). [Take $\sqrt{3} = 1.732$] [2]

— OR —

Find the length of the supporting cable running from observation station O to the top of Section B. [2]

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